

Introduction to Machine Learning

midsem1 - 2024-25



Semester

7



Year

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(a) Define a Well-Posed Learning Problem.

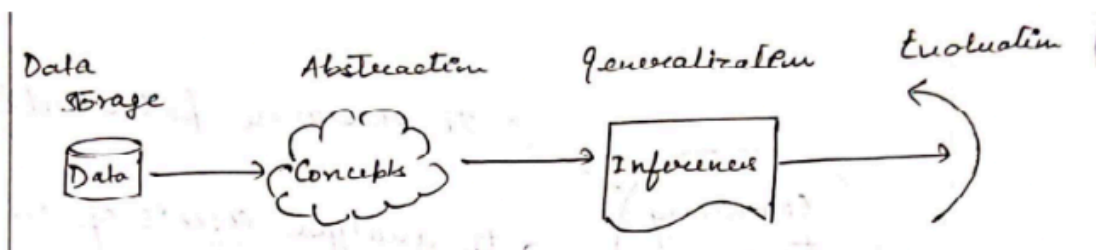
A computer program is said to learn from experience E in context to some task T and some performance measure P , if its performance on T , as was measured by P , upgrades with experience E .

Any problem can be segregated as well-posed learning problem if it has three traits -

- Task
- Performance Measure
- Experience

(b) What is Learning? State the components of Learning.

Learning is the process by which a system improves its performance on a task through experience.



Components of Learning

1. **Abstraction** : Simplifies the representation of a problem to make learning efficient.
2. **Evaluation** : Methods for measuring how good a hypothesis or model is.
3. **Generalization** : The ability of a model to perform well on **unseen data** drawn from the same distribution as the training data.
4. **Data Storage** : Storing and managing data for learning purposes efficiently.

(c) Define Inductive Bias in terms of Decision Tree Learning.

Inductive Bias refers to the set of assumptions a learning algorithm makes to predict outputs for unseen inputs.

In **Decision Tree Learning**, the inductive bias includes:

- Preferring shorter trees (with fewer nodes) over longer ones.
- Preferring trees where attributes that split the data well appear near the root.

This bias helps the algorithm **generalize** beyond the training data.

(d) Define Activation Function and state its types.

An **Activation Function** is a mathematical function applied to the output of a neuron to introduce **non-linearity** into the network, allowing it to learn complex patterns.

Types of Activation Functions:

1. **Linear Activation Function** – $f(x) = x$
 2. **Step Function** – Outputs 0 or 1 based on a threshold.
 3. **Sigmoid Function** – $f(x) = \frac{1}{1+e^{-x}}$ (Range: 0 to 1)
 4. **Tanh Function** – Range: (-1 to 1)
 5. **ReLU (Rectified Linear Unit)** – $f(x) = \max(0, x)$
 6. **Leaky ReLU** – Similar to ReLU but allows small negative values.
 7. **Softmax** – Converts values into probabilities.
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(e) Differentiate between a Single Layer and a Multilayer Neural Network.

Aspect	Single Layer Neural Network	Multilayer Neural Network
Structure	Has only an input layer and output layer	Has input, one or more hidden layers, and output layer
Complexity	Simple architecture	More complex architecture
Learning Ability	Can learn only linearly separable patterns	Can learn non-linear patterns
Example	Perceptron	Multi-Layer Perceptron (MLP), Deep Neural Network



(a) Differentiate between Data Science and Machine Learning

Aspect	Data Science	Machine Learning
Definition	A broad field focused on extracting insights from data using statistics, data analysis, and ML .	A subset of AI that enables machines to learn patterns from data and make predictions.
Scope	Covers data collection, cleaning, visualization, analysis, and modeling.	Focuses mainly on building and training models.
Techniques	Includes ML, statistical modeling, data mining, and visualization.	Uses algorithms like regression, SVM, decision trees, neural networks.
Tools	Python, R, SQL, Tableau, Hadoop.	TensorFlow, PyTorch, Scikit-learn.
Output	Insights and decision-making support.	Predictive models for automation and analytics.

(b) Explain Various Issues in Machine Learning

1. Overfitting & Underfitting

- Overfitting: Model performs well on training data but poorly on new data.
- Underfitting: Model is too simple and fails to capture patterns.

2. Insufficient Training Data

- Machine learning needs large, representative datasets for good performance.

3. High Dimensionality

- Too many features lead to the **curse of dimensionality**, making training slow and complex.

4. Noisy Data & Outliers

- Incorrect or inconsistent data reduces model accuracy.

5. Imbalanced Data

- When some classes dominate, the model becomes biased.

6. Feature Selection

- Choosing the right features is critical for performance.

7. Computational Complexity

- Large models require high processing power and memory.



(c) What is Artificial Neural Network (ANN) and its Relation to Biological Neural Network?

Definition:

An **Artificial Neural Network (ANN)** is a computational model inspired by the structure and functioning of the **biological brain**. It consists of interconnected nodes (neurons) organized in layers (input, hidden, output).

Relation to Biological Neural Network:

Biological Neural Network	Artificial Neural Network
Neurons process signals.	Nodes/units process inputs.
Synapses connect neurons.	Weights connect nodes.
Neuron fires if threshold met.	Activation function decides output.
Brain learns by adjusting synaptic strength.	ANN learns by adjusting weights using backpropagation.



(d) Explain Briefly Adaline Networks

Adaline (Adaptive Linear Neuron):

- A single-layer neural network proposed by **Widrow and Hoff**.
- Similar to Perceptron but uses **linear activation function** and **gradient descent** for weight updates.

Working:

1. Takes input features and calculates weighted sum.
2. Uses an **identity function** as activation.
3. Updates weights using **Least Mean Squares (LMS)** algorithm:

$$w_{new} = w_{old} + \eta(y - \hat{y})x$$

where η is the learning rate.

Characteristics:

- Used for linear classification problems.
- Foundation for more complex networks.



b) Types of Machine Learning with Examples

Machine Learning is broadly classified into **four main types** based on the nature of data and feedback:

1. Supervised Learning

- **Definition:**
The model is trained on a **labeled dataset**, meaning each training example has an input **X** and a known output **Y**.
 - **Goal:**
Learn a mapping function $f: X \rightarrow Y$ to predict the output for unseen inputs.
 - **Tasks:**
 - **Classification:** Output is categorical (e.g., Spam or Not Spam).
 - **Regression:** Output is continuous (e.g., Predicting house prices).
 - **Examples:**
 - Predicting email spam using labeled emails (spam/ham).
 - Weather prediction based on past temperature data.
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2. Unsupervised Learning

- **Definition:**
The model is trained on **unlabeled data**, so it must find **patterns, structure, or relationships** without explicit output labels.
- **Goal:**
Discover hidden structure in data.
- **Tasks:**
 - **Clustering:** Grouping similar data points.
 - **Association:** Finding rules that relate variables.
- **Examples:**
 - Customer segmentation in marketing.

- Grouping news articles by topic without labels.
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3. Semi-Supervised Learning

- **Definition:**
Uses a **small amount of labeled data** and a **large amount of unlabeled data** for training.
 - **Why:**
Labeling large datasets is expensive and time-consuming.
 - **Examples:**
 - Medical image classification where only some scans are labeled.
 - Speech recognition with limited labeled voice samples.
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4. Reinforcement Learning

- **Definition:**
The model (agent) learns by interacting with an environment and receiving **rewards** or **penalties** based on actions.
- **Goal:**
Maximize cumulative reward by choosing optimal actions.
- **Examples:**
 - Training a robot to walk.
 - AlphaGo learning to play Go by self-play.
 - Self-driving cars optimizing for safety and speed.



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